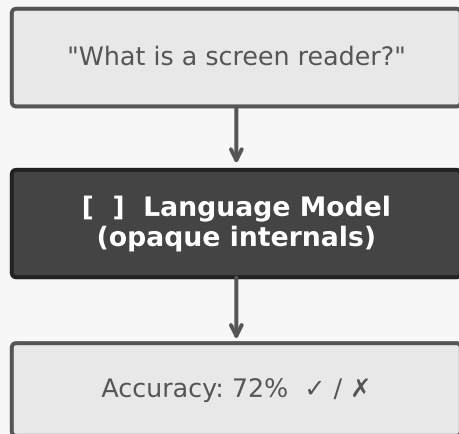


Behavioral Evaluation

(what models can do)



✗ **No insight into**

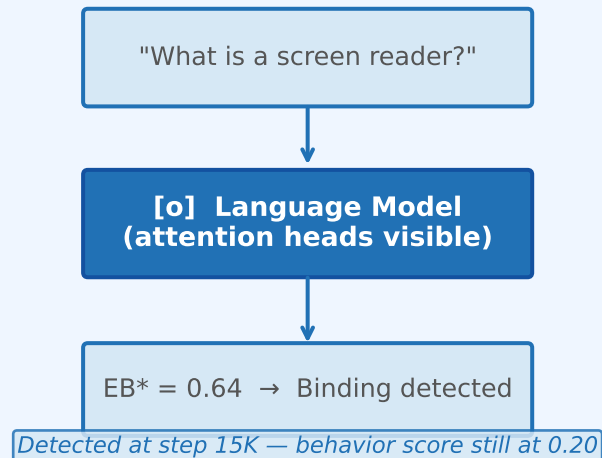
when internal representations form

✗ Expensive: run at every checkpoint

✗ Binary signal: pass or fail

Mechanistic Monitoring (EB*)

(how & when representations form)



✓ **Lightweight early-warning signal**

EB* peaks in the first 10% of training

✓ Forward-pass only — no auxiliary training

✓ Scales to 405B+, GQA-compatible, ~1 MB per pass